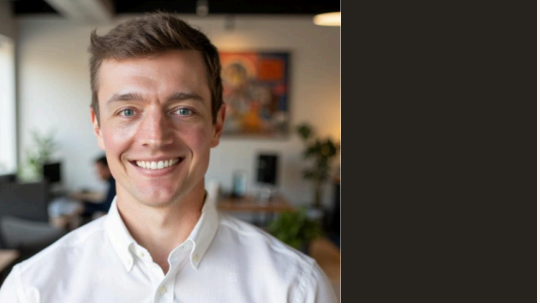


Lane Marsh

System Development Engineer
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LinkedIn

Engineer with a decade of experience in fleet reliability programs, data-driven maintenance analysis, and systems engineering. Experienced managing reliability metrics, identifying failure trends, and presenting actionable insights to engineering and operational stakeholders. Proficient in Python, SQL, and data pipeline development, with direct experience in aircraft health monitoring, and maintenance data systems.

Experience

Jun '24 - Now	**Amazon Prime Air - System Development Engineer** Maintain production data pipelines ingesting ~3TB of drone telemetry data daily for use by design engineers from 4,000 flights across 15 active sites. Tools are built with primarily Python and SQL in AWS environment (Lambda, RedShift, S3, CloudWatch) to support scaling up to 500 sites by the end of 2027. Developed and maintain drone maintenance data pipelines with ~200MB of daily traffic. Data pipelines support reliability engineering metrics and ML engineering to train categorization models. Built self-service dashboards that enable design engineering, supply chain and finance to analyze component performance. Serve as engineering on-call first responder during live drone operations: triage system alerts, investigate anomalies, and perform root cause analysis on drone performance issues.
Dec '19 - Jun '24	**Boeing - Systems Engineer** [KC-46 Program] Lead team of 5 engineers responsible for assessing hardware reliability performance and analyzing aircraft health management for internal and external purposes. Optimized analyses for teammates by developing ETL pipeline from PostgreSQL server to local environment using open source Python and SQL. Built business cases to address top fleet degrading systems around my team's analyses and presented progress to leadership on a bi-weekly cadence. [777/787 Program] Contributed to completion of the 787 Common Core System (CCS) System Safety Assessment by creating fault tree analyses related to power supply and network reliability. [P-8 Program] Designed and delivered sales pitches for a comparative business case analysis application. Automated generation of spare parts quotes based on analysis of customer failure reporting.
May '18 - Dec '19	**Alaska Airlines - Reliability Engineer** Developed python program utilizing naïve Bayes classifiers to categorize maintenance records by ATA code and automatically upload to TRAX. Developed Tableau dashboards analyzing flight and maintenance data. Presented monthly reliability metrics, worked with design engineers to address root cause of failure modes.
Jan '16 - May '18	**Boeing - Systems Engineer** [P-8 Program] Built Monte Carlo simulation tool using MS Excel and embedded VBA macros for modeling supply of replaceable fleet assets based on hardware reliability and the supporting supply chain. Presented quarterly executive briefs on readiness threats to the fleet.

Skills

Proficiencies

Python, SQL, Tableau, AWS (Redshift, QuickSight, Cloudwatch, Lambda, EC2, S3), Microsoft Office Suite, Atlassian Products (Confluence, Jira), Fault Tree Analysis

Hobbies

Tennis, Running, Guitar, Fly Fishing, Chess, Golf, Reading, F1

Education

Oregon State University

Jun 2023 - Bachelor Of Science in Computer Science

Dec 2015 - Bachelor Of Science in Industrial and Manufacturing Engineering